

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5	11	17	8
2	24	9	12	20	16
3	20	5	13	20	10
4	18	9	14	16	12
5	22	6	15	24	7
6	19	11	16	22	9
7	20	8	17	25	21
8	19	11	18	21	14
9	17	7	19	19	12
10	21	9	20	23	13

- What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?
- Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.
- Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

Student	With Imagery	Without Imagery
1	20	5
2	24	9
3	20	5
4	18	9
5	22	6
6	19	11
7	20	8
8	19	11
9	17	7
10	21	9
11	17	8
12	20	16
13	20	10
14	16	12
15	24	7
16	22	9
17	25	21
18	21	14
19	19	12
20	23	13

Stu	Diff	Rank	Sign.
1	15	17	+
2	15	17	+
3	15	17	+
4	9	8.5	+
5	16	19	+
6	8	6.5	+
7	12	13.5	+
8	8	6.5	+
9	10	11	+
10	12	13.5	+
11	9	8.5	+
12	4	2	+
13	12	13.5	+
14	4	2	+
15	17	20	+
16	13	15	+
17	4	2	+
18	7	4.5	+
19	7	4.5	+
20	10	11	+

(a). paired t-Test

Wilcoxon Signed - Rank test

Sign Test

t-test assumption

- data are paired
- the paired differences are independent
- the differences follow a normal distribution

The differences unevenly distributed $\rightarrow$ the third assumption is not satisfied.

(b)

Sign: $H_0: p=0.5$, $H_a: p \neq 0.5$

$\oplus: 20$, $\ominus: 0$

$X = 20$

$p\text{-value} = 2P(X \geq 20) = 0$

$\because p\text{-value} < 0.05 \Rightarrow \text{reject } H_0$

There is sufficient evidence to say that recall performance are difference

Wilcoxon Signed - Rank

$T^+ = 20$, $T^- = 0$, $T = 0$

$\alpha = 0.05$

$RR = \{T \mid T \leq 52\}$ $\because 0 \leq 52$
 $\therefore \text{Reject } H_0$

There's a significant difference

(c).

Yes, both test lead to same conclusion.